

DTH TV: What's Available

DTH set-top box and dish antenna are today capable of delivering much more than playing, pausing, rewinding and recording live TV. Ensure you pick the right system, steering clear of misleading claims and technical jargon



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The larger-than-life game of cricket, worshipped throughout our nation, is best enjoyed in a stadium as a live sporting event." Defying this belief, big screens running high-definition (HD) content via direct-to-home (DTH) TV technology can certainly give you all the thrills and action you crave for, right in your living room. Thanks to the DTH revolution in India, it is most likely that you have already subscribed to a DTH service for your entertainment needs. But if your long-term relationship with that neighbourhood *cablewala* gives you nothing more than sub-standard analogue-quality TV services, it's time to switch to something better.

To avail DTH TV service, you need a set-top box (STB) and a small dish antenna, which is generally erected on the rooftop or at a height where broadcast signals can be received directly from a satellite. The received signals

are then decoded to viewable digital content via the subscriber's STB that is connected to a television.

Unlike a few years ago when India's first DTH service Dish TV served as the only bridge in town, a number of operators such as Reliance Digital TV, Tata Sky, Airtel Digital TV and Videcon D2H now offer services ranging from HD channels to interactive services to 3D capability. So before you buy a DTH TV system, here is a checklist to help you pick the best system for you.

Compression technology

In India, DTH operators use two kinds of compression technologies: MPEG-2 and MPEG-4. The early players in the market (Dish TV and Tata Sky) provide their services using MPEG-2 compression, whereas relatively newer entrants like Reliance Digital TV, Videcon D2H and Airtel Digital TV have brought MPEG-4 compression to the market. Operators use marketing tactics to promote MPEG-4 as superior to MPEG-2, which may not always be the case.

Every DTH service provider rents transponders on a satellite that collects and transmits TV signals in digital format. These signals are transmitted to dish antennae connected to a set-top box. The transponders have a specific bandwidth. Their channel capacity is derived from the bit rate of individual channels. If a given transponder has a bandwidth of 128 Mbps, and each channel is encoded at a bit rate of 2 Mbps, it can hold 64 channels.

Technically, MPEG-4 offers bet-

ter compression than MPEG-2. So several DTH service providers choose this format in order to narrow down an individual channel's bit rate. This increases the scope of accommodating more channels into a single transponder's bandwidth. For example, if the DTH operator reduces the bit rate to 1.6 Mbps, the transponder can hold 80 channels.

However, in case of video encoding, loss in bit rate leads to a relative loss in the overall video quality too. In a nutshell, compression technologies offer a choice to DTH operators. They could either pass on the quality benefit to the consumers or pump in more channels and eventually save costs incurred on renting transponders.

Channels offered

Popcorn, cans of chilled beer, a large LCD TV—the necessary ingredients to set the home on fire during the broadcast of an interesting event like India-Pakistan cricket match—will all go down the drain if the event isn't being broadcast on your favourite sports channel. Aggressive revenue models have forced content producers to often move from

TABLE I
Set-up Costs* for New Connections

Set-up	Cost (Rs)
Airtel Digital TV	1590
Dish TV	1440
Reliance	1540
Tata Sky	1415
Videcon D2H	1540

*The cost includes a standard set-top box with some companies offering free subscription for a month or a regular channels package. If you prefer HD, you will need to spend a few extra bucks for an HD set-top box.